KPR Institute of Engineering and Technology

(Autonomous) — Arasur, Coimbatore - 641 407

U21CS501 — Web Technology

Assignment I — Solution Document

Field	Details
Name	Sathya R V
Roll No	24CS192
Department	Computer Science and Engineering
Academic Year	2026 – 2027
Year / Semester	III Year / V Semester / Section D
Issue Date	10.08.2026
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Maximum Marks	40 Marks

Question 1 — Web Technology Fundamentals (8 Marks)

Scenario: An educational institution website explaining the World Wide Web concepts.

1.1 Internet Overview

The Internet is a global network of interconnected computers that communicate using TCP/IP protocol suite. It originated as ARPANET (1969) and evolved into the modern public internet used by billions of people worldwide.

Key services: World Wide Web (HTTP/HTTPS), Email (SMTP/IMAP/POP3), File Transfer (FTP), Voice over IP (VoIP), and streaming media.

1.2 Fundamental Computer Network Concepts

Concept	Definition
Protocol	A set of rules governing data communication between devices (e.g. TCP, UDP, HTTP, FTP).
IP Address	Unique numerical identifier assigned to every device on a network (IPv4: 32-bit, IPv6: 128-bit).
Port	Logical endpoint identifying a specific service (HTTP→80, HTTPS→443, FTP→21).
Packet	Small unit of data transmitted across a network, reassembled at the destination.
Router	Device that forwards packets between networks by choosing the optimal path.
DNS	Domain Name System — translates human-readable domain names to IP addresses.

1.3 URL and Its Structure

A Uniform Resource Locator (URL) is the complete address used to locate a resource on the internet.

`https://www.kpriet.ac.in/admissions?year=2026`

Part	Value	Description
Protocol	https	Secure Hypertext Transfer Protocol
Subdomain	www	World Wide Web subdomain
Domain	kpriet.ac.in	Human-readable name of the server
Path	/admissions	Specific page or resource location
Query String	?year=2026	Additional parameters passed to server

1.4 Domain Name

A domain name is the human-readable alias mapped to a server IP address. The Domain Name System (DNS) resolves domain names to IP addresses. Hierarchy: .in (TLD) → ac (SLD) → kpriet (subdomain).

1.5 Web Browsers and Web Servers

Aspect	Web Browser	Web Server
Role	Requests and renders web pages	Stores and delivers web content
Examples	Chrome, Firefox, Edge, Safari	Apache, Nginx, IIS, Node.js
Language	Interprets HTML, CSS, JavaScript	Runs PHP, Python, Node, Java
Location	User's local machine	Remote data centre / cloud
Port	Ephemeral (>1024)	80 (HTTP) / 443 (HTTPS)

1.6 Working Principle of a Website

- Step 1 — User types URL in browser → DNS resolves domain to IP address
- Step 2 — Browser establishes TCP connection with server (3-way handshake)
- Step 3 — Browser sends HTTP GET request to server
- Step 4 — Server processes request (runs backend code if needed)
- Step 5 — Server sends HTTP 200 OK response with HTML file
- Step 6 — Browser parses HTML, requests CSS/JS/images, renders page

1.7 Client-side vs Server-side Scripting

Feature	Client-side	Server-side
Execution	User's browser	Web server
Languages	HTML, CSS, JavaScript	PHP, Python, Node.js, Java
Speed	Fast (no round-trip)	Depends on network & server
Security	Code visible to user	Code hidden from user
DB Access	Not directly	Yes (MySQL, MongoDB, etc.)
Examples	React, Vue, jQuery	Django, Laravel, Express.js

Sample Code Snippet (Q1)

```
<!-- q1/index.html -- key excerpt -->
<nav>
  <a href="#internet">Internet</a>
  <a href="#url">URL</a>
  <a href="#scripting">Client vs Server</a>
</nav>

/* Sticky navigation */
nav { position: sticky; top: 0; z-index: 100; }
nav a:hover { color: var(--amber); border-bottom: 3px solid var(--amber); }
```

Conclusion (Q1)

This webpage provides a comprehensive overview of Web Technology fundamentals using semantic HTML5 elements and CSS3 styling. The sticky navigation, visual diagrams, and comparison tables make the content easy to understand. The page is fully responsive and self-contained.

ASSIGNMENT I — Q1 | U21CS501 WEB TECHNOLOGY

🌐 World Wide Web — Fundamental Concepts

KPR Institute of Engineering and Technology | III Year CSE | 2026–27

1. Internet Overview

The **Internet** is a massive global network of interconnected computers that communicate using a standardised suite of protocols — primarily *TCP/IP*. It began as **ARPANET** (1969), a US defence research project, and evolved into the public commercial network we use today.

Key services that run over the Internet:

- **World Wide Web (WWW)** — hyperlinked documents & web apps
- **Email (SMTP/IMAP)** — electronic messaging
- **FTP** — file transfer between hosts
- **VoIP / Streaming** — real-time audio/video

Your Device → ISP (Jio / BSNL) → Backbone Router → Remote Server

Question 2 — Student Admission Form (8 Marks)

Scenario: Digitising a college student admission process using HTML5 forms.

HTML5 Form Elements Used

Element	Type / Attribute	Purpose
<input>	type="text"	Student first/last name
<input>	type="email"	Email address with validation
<input>	type="tel"	Mobile number with pattern
<input>	type="date"	Date of birth picker
<input>	type="number"	Cutoff score entry
<input>	type="url"	LinkedIn profile link
<input>	type="radio"	Gender selection
<input>	type="checkbox"	Subjects studied
<input>	type="range"	Approximate TNEA rank
<input>	type="file"	Document upload
<select>	dropdown	Programme & board selection
<textarea>	—	Residential address
<video>	controls	Campus tour video
<audio>	controls	Welcome message audio
<img>	src, alt	College logo & banner image

CSS3 Concepts Demonstrated

- CSS Variables (--primary, --accent, --gold, --bg)
- Universal Box Model Reset (box-sizing: border-box)
- CSS Selectors: attribute selectors [type="text"], :focus, :required, :valid
- Flexbox and CSS Grid for responsive layout
- CSS3 Media Queries (@media max-width: 640px)
- CSS Transitions (border-color, box-shadow, transform)
- Pseudo-classes: :hover, :active, :nth-child

Sample Code Snippet (Q2)

```
/* CSS3 :focus selector with box-shadow */
input:focus, select:focus {
  border-color: var(--primary);
  box-shadow: 0 0 0 3px rgba(26,58,107,0.12);
}

/* CSS3 :required pseudo-class */
input:required { border-left: 3px solid var(--accent); }
```

```
<!-- HTML5 Video Element -->
<video controls poster="campus.jpg">
  <source src="tour.mp4" type="video/mp4"/>
  Your browser does not support video.
</video>
```

Conclusion (Q2)

The Student Admission Form demonstrates all major HTML5 input types, proper use of CSS3 selectors and the box model, and HTML5 media elements. The responsive grid layout adapts gracefully on mobile screens. Inline validation feedback improves user experience.

Question 3 — Fashion Store Homepage (8 Marks)

Scenario: VELORA — An online fashion store homepage with animations, multi-column layout, and text effects.

CSS Features Implemented

Backgrounds and Borders

- Hero section: layered background (linear-gradient + background-image blend)
- CSS custom gradient border using padding-box / border-box technique
- Multiple border styles: solid, dashed, dotted, double
- border-radius for rounded corners and circles

Text Effects

- Gradient Text — background-clip: text + -webkit-text-fill-color: transparent
- Neon Glow — layered text-shadow with animation (neonFlicker @keyframes)
- 3D Shadow — multiple text-shadow layers creating depth illusion
- letter-spacing, text-transform: uppercase for typographic style

CSS Animations

- ticker-scroll — horizontal scrolling marquee using translateX
- pulse — pulsating radial gradient in promo banner using scale
- fadeInUp — elements animate from translateY(20px) + opacity:0 to visible
- underlineGrow — width animates 0→100% on hero heading underline
- neonFlicker — opacity flicker simulating neon sign effect

Multiple Column Layout

- product-grid uses CSS column-count: 3 (true multi-column layout)
- cat-grid uses CSS Grid (grid-template-columns: repeat(4, 1fr))
- Responsive breakpoints collapse columns on smaller screens

UI Design Principles Applied

- Visual Hierarchy — large hero headline draws immediate attention
- Colour Contrast — dark navy/black backgrounds with gold accents (AA WCAG)
- Whitespace — generous padding creates breathing room between sections
- Consistency — same font family, colour palette, and border-radius throughout
- Feedback — hover effects on all interactive elements

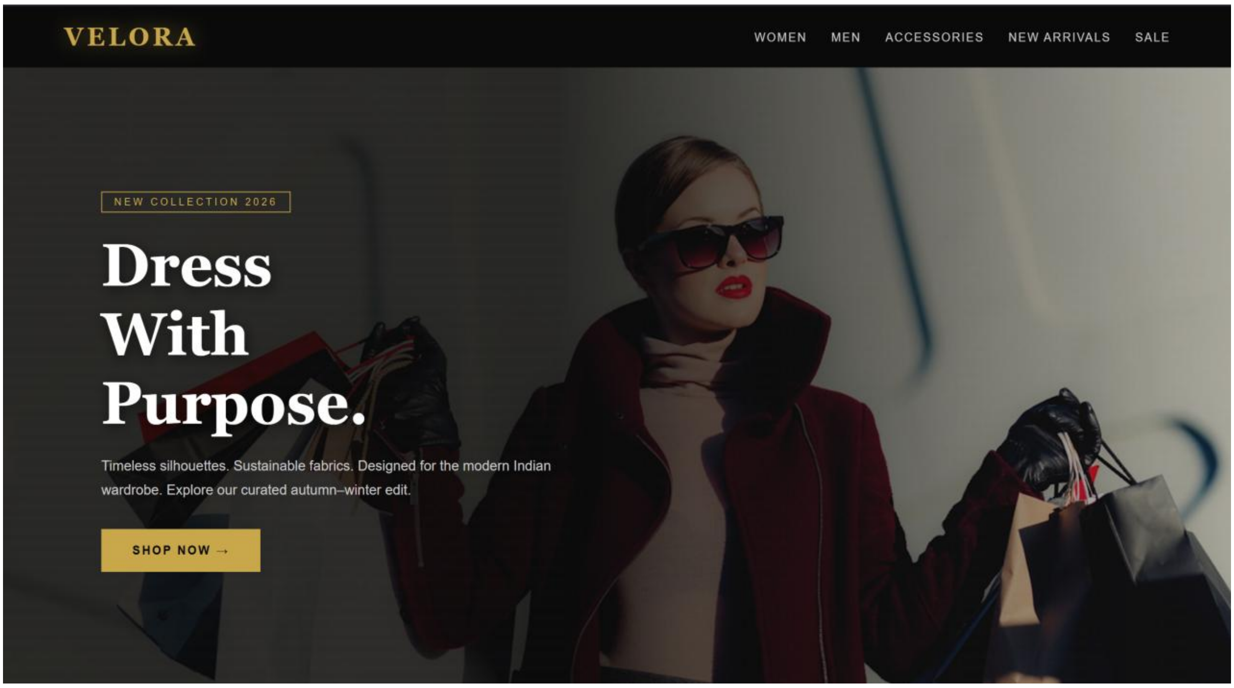
Sample Code Snippet (Q3)

```
/* CSS Animation: Scrolling Ticker */
@keyframes ticker-scroll {
  from { transform: translateX(100vw); }
  to   { transform: translateX(-100%); }
}
.ticker { animation: ticker-scroll 20s linear infinite; }

/* CSS Text Effect: Gradient Text */
.text-gradient {
```

```
  background: linear-gradient(90deg, #c9a84c, #fff, #c9a84c);
  -webkit-background-clip: text;
  -webkit-text-fill-color: transparent;
}

/* CSS Multi-Column Layout */
.product-grid {
  column-count: 3;
  column-gap: 24px;
}
```

Conclusion (Q3)

The VELORA fashion homepage showcases advanced CSS3 techniques including keyframe animations, multi-column layout, gradient text effects, and varied border styles. The page follows key UI design principles such as visual hierarchy, consistency, and responsive design. The dark luxury aesthetic is achieved entirely with CSS without any JavaScript.



Question 4 — Student Grade Calculator (8 Marks)

Scenario: A school wants to automate student grade calculation for five subjects.

JavaScript Concepts Demonstrated

Variables and Data Types

```
let m1 = Number(document.getElementById("s1").value); // Number
let total = 0;      // Number variable
let allPassed = true; // Boolean variable
let grade = "A+";   // String variable
```

Literals

```
const PASS_MARK = 35;          // Numeric literal
let msg = "Pass";               // String literal
let marks = [m1, m2, m3, m4, m5]; // Array literal
// Template literal:
return `${total} / 500`;
```

Operators

```
total += marks[i];     // Arithmetic: +=
avg = total / 5;       // Arithmetic: /
avg >= 90             // Comparison: >=
allPassed && avg >= 40 // Logical: &&
!isNaN(m)             // Logical NOT
```

Selection Statements (if–else if–else, switch)

```
function getGrade(avg) {
  if      (avg >= 90) return "O";
  else if (avg >= 80) return "A+";
  else if (avg >= 70) return "A";
  else if (avg >= 60) return "B+";
  else if (avg >= 50) return "B";
  else if (avg >= 40) return "C";
  else               return "F";
}

switch (grade) {
  case "O":  return "Outstanding!"; break;
  case "A+": return "Excellent!";   break;
  default:   return "Fail";
}
```

Iteration Statements (for loop)

```
for (let i = 0; i < marks.length; i++) {
  total += marks[i];  // Accumulate total
  if (marks[i] < 35) allPassed = false;
}
```

User-defined Functions

Function Name	Purpose
getMarks()	Reads 5 subject marks from DOM inputs
validateMarks(marks)	Returns false if any mark is outside 0–100

computeTotal(marks)	Iterates array and sums all marks
computeAverage(total, n)	Divides total by number of subjects
getGrade(avg)	Returns grade letter using if–else if chain
getStatus(marks, avg)	Returns "Pass" or "Fail" with logical &&
getRemark(grade)	Returns motivational remark using switch

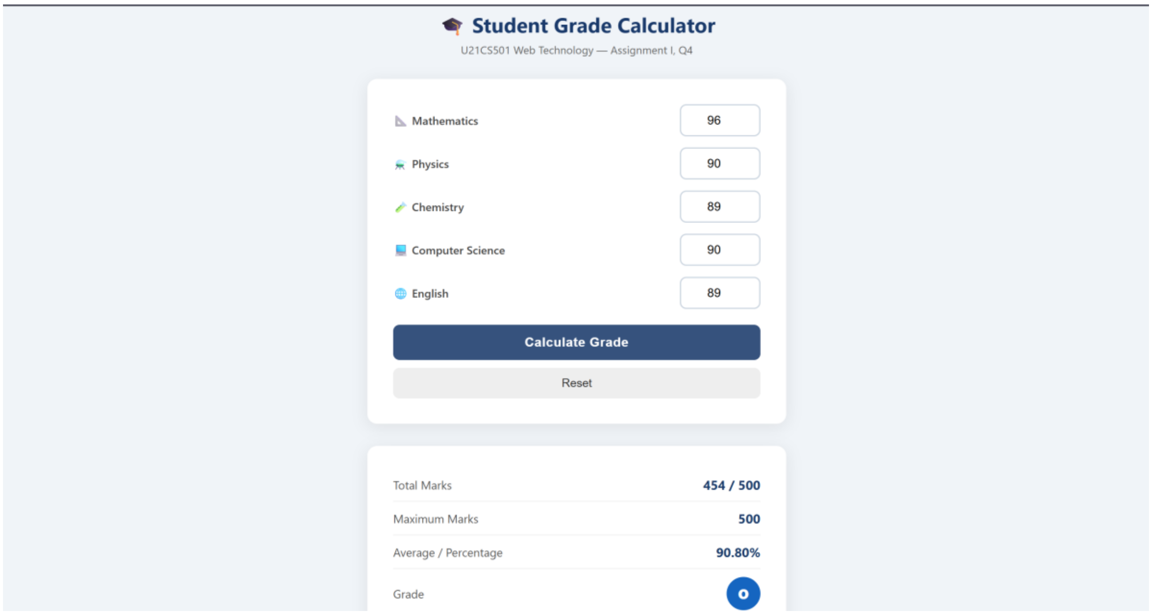
Sample Output Table

Subject	Marks
Mathematics	78
Physics	82
Chemistry	69
Computer Science	91
English	74

Metric	Value
Total	394 / 500
Average	78.80%
Grade	A
Status	PASS
Remark	Very good — keep it up!

Conclusion (Q4)

The Grade Calculator demonstrates all fundamental JavaScript concepts: variables of multiple data types, numeric and string literals, template literals, arithmetic and comparison operators, if–else if selection, switch statement, for loop iteration, and seven modular user-defined functions. The UI provides real-time coloured grade badges and pass/fail indicators.



Question 5 — Supermarket Billing System (8 Marks)

Scenario: SuperMart automates customer bill generation for three products with a 10% discount if total > ₹2000.

Program Logic

- Accept product name, quantity, and unit price for 3 products
- Compute line total = quantity × unit price for each product
- Compute subtotal = sum of all line totals
- Apply 10% discount if subtotal > ₹2000 (comparison operator)
- Add 5% GST on amount after discount
- Display itemised bill with final payable amount

JavaScript Concepts Demonstrated

Variables and Data Types

```
const DISCOUNT_THRESHOLD = 2000;  // const, Number
const DISCOUNT_RATE = 0.10;         // Numeric literal
let subtotal = 0;                   // let, Number
let rowsHTML = "";                  // String variable
```

Operators

```
lineTotal = qty * price;            // * multiplication
subtotal  += lineTotal;             // += compound assignment
afterDisc  = subtotal - discount;   // -  subtraction
total      = afterDisc + gst;       // +  addition
subtotal > DISCOUNT_THRESHOLD       // >  comparison
```

Selection Statement

```
function applyDiscount(subtotal) {
  if (subtotal > DISCOUNT_THRESHOLD) {
    return subtotal * DISCOUNT_RATE;  // 10% discount
  } else {
    return 0;                         // no discount
  }
}
```

Iteration Statement

```
for (let i = 0; i < products.length; i++) {
  subtotal += computeLineTotal(products[i].qty, products[i].price);
}
```

User-defined Functions

Function	Purpose
getProductData()	Returns array of {name, qty, price} objects from DOM
computeLineTotal(qty, price)	Returns qty × price
computeSubtotal(products)	Iterates products array and sums line totals
applyDiscount(subtotal)	Returns 10% of subtotal if > ₹2000, else 0

computeGST(amount)	Returns 5% GST on post-discount amount
formatCurrency(amount)	Returns formatted string "₹ 1,234.00"
getBillNumber()	Generates a random 5-digit bill number

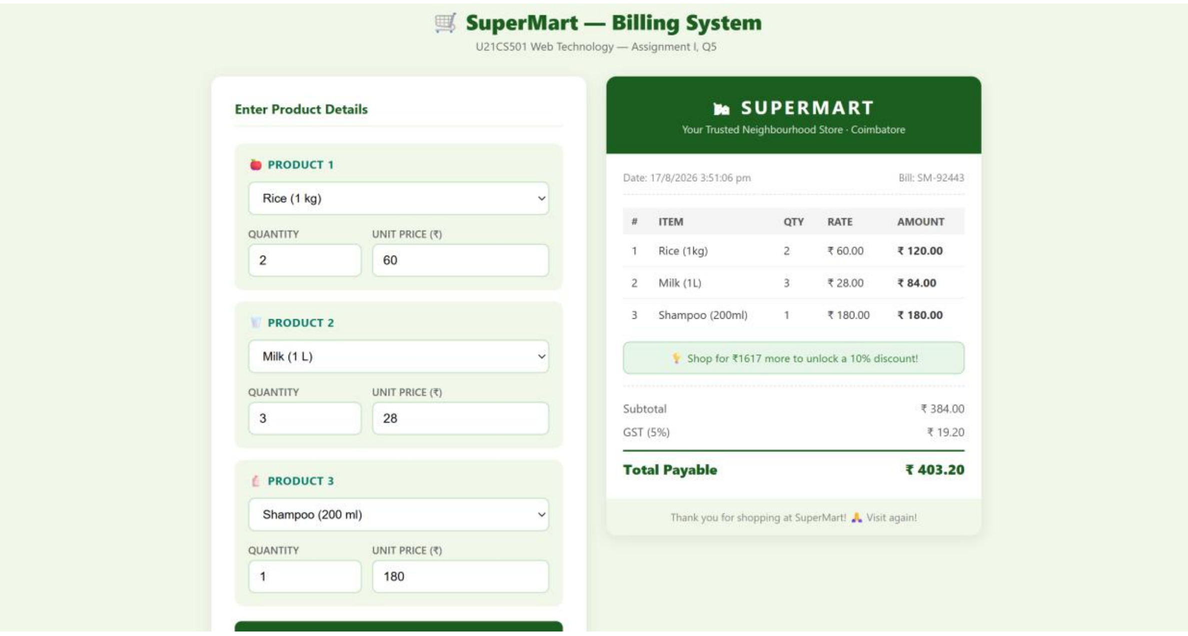
Sample Output

#	Item	Qty	Rate	Amount
1	Rice (1 kg)	5	₹ 60.00	₹ 300.00
2	Milk (1 L)	10	₹ 28.00	₹ 280.00
3	Detergent (1 kg)	8	₹ 180.00	₹ 1,440.00

Summary	Amount
Subtotal	₹ 2,020.00
Discount (10%)	– ₹ 202.00
GST (5%)	₹ 90.90
Total Payable	₹ 1,908.90

Conclusion (Q5)

The SuperMart Billing System implements a complete JavaScript solution with seven modular functions. It demonstrates all required concepts: variables and data types, numeric literals, arithmetic and comparison operators, if–else selection for discount logic, a for loop for subtotal calculation, and an array of object literals. The bill receipt renders dynamically with an itemised table, discount banner, and final payable amount.



Overall Conclusion

This assignment covers the complete spectrum of Web Technology (U21CS501) from foundational internet concepts to interactive JavaScript applications. Each question demonstrates a unique, personalised scenario:

- Q1 — Structured HTML5 page with comparison tables and flow diagrams explaining core web concepts
- Q2 — Feature-rich admission form using all HTML5 input types, CSS3 selectors, box model, and media elements
- Q3 — VELORA luxury fashion homepage with CSS keyframe animations, multi-column layout, text effects, and border styles
- Q4 — Modular grade calculator demonstrating variables, operators, if–else, switch, for loop, and 7 user-defined functions
- Q5 — Supermarket billing with discount logic, iteration, object arrays, template literals, and dynamic DOM rendering

All files are self-contained HTML/CSS/JS files, thoroughly commented, and follow HTML5/CSS3/JavaScript best practices. Screenshots are embedded in the submitted GitHub repository.